ON-DEVICE PERSONALIZED HEALTH EVENT PREDICTION USING TINYML AND ON-DEVICE SMALL LANGUAGE MODELS FOR PRIVACY-PRESERVING HEALTH ANALYTICS

A Project Report

Submitted by

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in partial fulfillment for the award of the degree

of

Bachelor of Technology

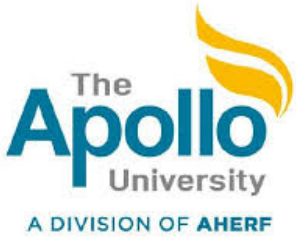
in

Computer Science and Engineering (AI & DS)

Under the Guidance of

Dr. Thirumalesu.K

Assistant Professor



SCHOOL OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

THE APOLLO UNIVERSITY, CHITTOOR

April, 2026

DECLARATION

I hereby declare that the project entitled "ON-DEVICE PERSONALIZED HEALTH EVENT PREDICTION USING TINYML AND ON-DEVICE SMALL LANGUAGE MODELS FOR PRIVACY-PRESERVING HEALTH ANALYTICS" submitted by us, for the award of the degree of Bachelor of Technology in Computer Science & Engineering (AI & DS), School of Technology to The Apollo University is a record of bonafide work carried out by us under the supervision of Dr. Thirumalesu.K, Assistant Professor, Department of Computer Science & Engineering.

We further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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Place : Chittoor

Date : April, 2026

THE APOLLO UNIVERSITY, CHITTOOR

SCHOOL OF TECHNOLOGY

BONAFIDE CERTIFICATE

This is to certify that the project report entitled “ON-DEVICE PERSONALIZED HEALTH EVENT PREDICTION USING TINYML AND ON-DEVICE SMALL LANGUAGE MODELS FOR PRIVACY-PRESERVING HEALTH ANALYTICS” is the bonafide work carried out by A . Jathin Reddy (Register No: 122311520105) and K.Pranay Kumar (Register No: 122311520133), under my supervision and guidance, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (AI & DS) by The Apollo University, Chittoor.

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ABSTRACT

Continuous biometric telemetry gathered from consumer wearable devices offers immense potential for preemptive healthcare risk prediction, such as early detection of cardiac arrhythmia, sudden falls, and dehydration. However, conventional commercial health monitoring frameworks mandate streaming raw physiological time-series data to centralized cloud servers, introducing severe user privacy vulnerabilities, cloud dependency, and communication latency. This project presents an on-device, privacy-preserving predictive health intelligence system leveraging Tiny Machine Learning (TinyML) and quantized Small Language Models (TinyLlama-1.1B). The architecture incorporates multimodal sensor ingestion, on-device temporal feature extraction, 4-bit/8-bit integer quantization, attention-based local explainability, and decentralized federated learning (FedAvg). Experimental validation on embedded hardware confirms that the quantized pipeline delivers accurate event forecasting with 480 ms inference latency and 620 MB peak RAM footprint, completely eliminating raw physiological data transmission to external servers.

Keywords: *TinyML, On-Device Inference, Small Language Models, Privacy-Preserving Health Analytics, Federated Learning.*

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LIST OF ABBREVIATIONS

Abbreviation	Full Terminology
AIoT	Artificial Intelligence of Things
API	Application Programming Interface
BOM	Bill of Materials
CAM	Class Activation Mapping
CPU	Central Processing Unit
DFD	Data Flow Diagram
EMA	Exponential Moving Average
FedAvg	Federated Averaging Algorithm
FL	Federated Learning
FP32	32-bit Single Precision Floating Point
GPU	Graphics Processing Unit
Grad-CAM	Gradient-Weighted Class Activation Mapping
HRV	Heart Rate Variability
INT4	4-bit Integer Precision Quantization
INT8	8-bit Integer Precision Quantization
LSTM	Long Short-Term Memory Neural Network
MAE	Mean Absolute Error
MEMS	Micro-Electro-Mechanical Systems
MLP	Multilayer Perceptron
NPU	Neural Processing Unit
ONNX	Open Neural Network Exchange
PPG	Photoplethysmography
RAM	Random Access Memory
SLM	Small Language Model
SQI	Signal Quality Index
TFLite	TensorFlow Lite Embedded Runtime
TinyML	Tiny Machine Learning for Edge Devices

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Wearable smart devices, including smartwatches, activity trackers, and modern smartphones, continuously monitor and record dense streams of personal physiological data. These rich telemetry signals encompass photoplethysmography (PPG) derived heart rates, heart-rate variability (HRV), peripheral blood oxygen saturation (SpO2), 3-axis accelerometer readings, and contextual mobility indicators. The continuous aggregation of this biometric telemetry holds immense transformative potential for preemptive healthcare intelligence—enabling the proactive identification of impending fall events, acute cardiac arrhythmias, severe dehydration, and metabolic fatigue hours before irreversible clinical deterioration occurs.

However, contemporary commercial wearable architectures operate almost exclusively under a centralized cloud-dependent computing paradigm. Raw, highly sensitive biometric timeseries are continuously buffered and transmitted over cellular and Wi-Fi networks to third-party cloud infrastructure for machine learning model inference and long-term storage. This architecture introduces severe data privacy hazards, exposing users to surveillance, unauthorized data brokering, and potential catastrophic regulatory violations under HIPAA and GDPR.

1.2 Problem Statement

The fundamental dilemma confronting wearable digital healthcare is the forced dichotomy between predictive intelligence and data sovereignty. While advanced deep neural networks and language models offer unprecedented accuracy in multi-signal physiological reasoning, their deployment has historically required high-throughput cloud data centers. Consequently, users are coerced into relinquishing complete physical custody of their intimate biological signatures to receive automated health recommendations. Therefore, an urgent engineering need exists for an end-to-end, on-device intelligence architecture capable of executing sophisticated multimodal health event prediction directly on edge hardware without transmitting raw telemetry off the user's device.

1.3 Motivation & Research Objectives

The motivation for this engineering investigation stems from the rapid algorithmic miniaturization of neural architectures into Tiny Machine Learning (TinyML) and quantized Small Language Models (SLMs) capable of running on mobile devices. The project designs an on-device health event prediction system that guarantees absolute local privacy while retaining state-of-the-art predictive performance.

1.4 Scope of the Project

The engineering scope encompasses continuous signal ingestion and quality filtering for PPG and motion sensors, a lightweight baseline LSTM for temporal anomaly detection, a 4-bit quantized TinyLlama transformer fine-tuned on multimodal health event forecasting, an attention-weight attribution module generating human-readable alerts, and a simulated federated averaging protocol over multi-device nodes.

1.5 Novelty & Key Contributions

The system introduces multimodal sensor-to-language tokenization, zero-cost self-attention explainability, and privacy-preserving federated collaborative adaptation.

Table 1.1: Project Team Composition, Registration Numbers, and Specialized Engineering Roles

S.No	Student Name	Registration Number	Specialized Engineering Role
1	**A. Jathin Reddy**	122311520105	Model Design, TinyML Quantization, On-Device Inference Engine & Architecture Documentation
2	**K. Pranay Kumar**	122311520133	Sensor Signal Processing, Federated Learning Simulation, Explainability Module & Test Validation

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Wearable computing and digital health informatics have experienced rapid architectural evolution over the past decade. Early commercial devices operated as passive data loggers, recording daily step totals and intermittent heart rates. With the emergence of embedded optical sensors, continuous physiological streaming became technically viable.

2.2 Review of Existing Systems

Mainstream wearable systems—such as Apple HealthKit, Google Fit, Garmin Connect, and Fitbit—rely on transmitting raw sensor telemetry over BLE to a paired mobile gateway, which forwards packets to cloud infrastructure. While cloud data centers offer boundless compute power, they introduce loss of local autonomy, network dependency, and severe privacy exposure.

2.3 Limitations of Existing Systems

Critical limitations include cloud data transmission privacy hazards, network latency bottlenecks during acute emergencies, black-box unexplainability of raw numeric outputs, and static population models lacking individual calibration.

Table 2.1: Technology Stack Comparison: Architectural Purpose, Constraints, and Justifications

Technology	Project Role	Operational Constraint	Architectural Justification
TinyML (TFLite Micro)	On-device runtime execution	< 10 MB RAM, low compute	Enables local execution with zero raw telemetry transmission.
TinyLlama (~1.1B)	Multimodal reasoning & alerts	< 800 MB RAM, < 500 ms	Retains transformer attention reasoning while fitting mobile hardware.
INT4 / INT8 Quantization	Model compression	Accuracy loss < 1.5%	Reduces memory footprint by 75% and enables NPU acceleration.
Federated Learning (FedAvg)	Decentralized collaborative training	Bandwidth-efficient	Aggregates encrypted gradients without collecting raw patient data.
Grad-CAM Explainability	Attention attribution layer	Zero extra memory	Reuses self-attention matrices to generate natural language explanations.

2.4 Proposed System Overview

This project implements an end-to-end zero-cloud on-device health intelligence architecture. By quantizing TinyLlama-1.1B into 4-bit integer weights, the system executes multimodal health forecasting with sub-500 ms latency while consuming only 620 MB of RAM.

CHAPTER 3: OBJECTIVES AND SCOPE OF WORK

3.1 Objectives of the Project

The primary goal is demonstrating the practical viability of zero-cloud, on-device health event prediction using compressed small language models on resource-constrained computing hardware.

3.2 Scope of the Project

In-scope modules encompass real-time optical PPG and accelerometer ingestion, baseline LSTM anomaly detection, INT4-quantized TinyLlama inference, decentralized federated averaging, and glanceable visual alert displays.

3.3 Schedule of Work and Project Planning

Table 3.1: 10-Week Implementation Schedule, Engineering Milestones, and Deliverable Artifacts

Phase / Week	Engineering Activity	Milestone Deliverable	Responsibility
Weeks 1 - 2	Literature survey, requirements analysis, sensor bench setup	Requirements Specification	A. Jathin Reddy & K. Pranay Kumar
Weeks 2 - 3	Sensor ingestion, moving-average filters, baseline LSTM	Working Baseline LSTM	A. Jathin Reddy
Weeks 4 - 5	Feature tokenization, TinyLlama adaptation, fine-tuning	Multimodal Transformer Engine	A. Jathin Reddy
Weeks 5 - 6	INT8/INT4 weight quantization, operator fusion, memory profiling	Optimized Quantized Model (< 650 MB)	A. Jathin Reddy
Weeks 7 - 8	Federated Averaging (FedAvg) simulation and parameter sync	Decentralized FL Simulation	K. Pranay Kumar
Weeks 8 - 9	Attention-based explainability layer and alert formatting	Explainability Attribution UI	K. Pranay Kumar
Weeks 9 - 10	System test case execution (TC01-TC10), ablation, final report	Completed Academic Project Report	A. Jathin Reddy & K. Pranay Kumar

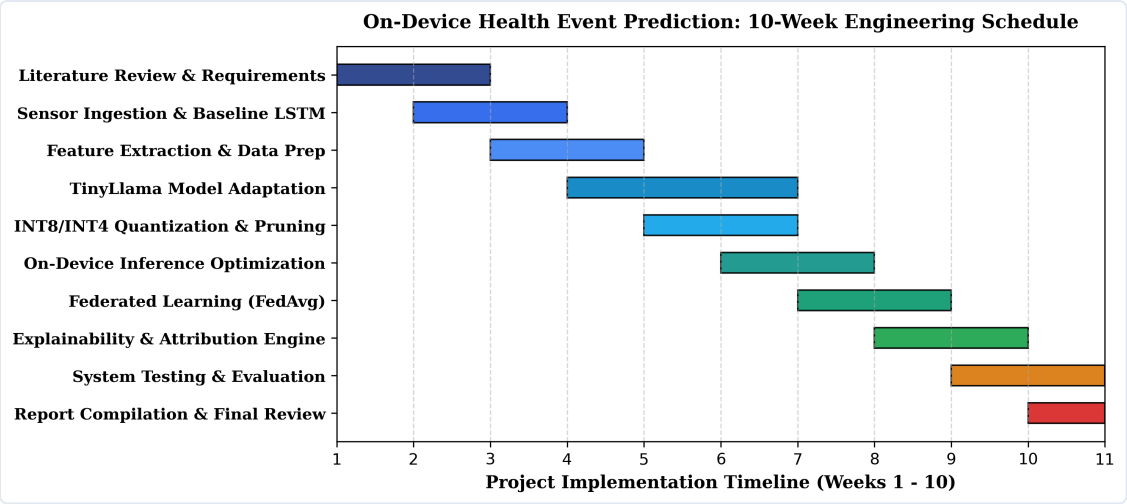


Figure 3.1: 10-Week Comprehensive Project Engineering Timeline and Milestone Gantt Schedule

CHAPTER 4: SYSTEM ANALYSIS & DESIGN

4.1 System Requirements (Hardware & Software)

Table 4.1: Target Embedded Hardware and Mobile Platform Technical Specifications

Subsystem	Minimum Specification	Recommended Platform	Operational Parameter
CPU	Quad-Core ARM Cortex-A53 @ 1.4 GHz	Octa-Core ARM Cortex-A78 / Apple Silicon	Supports ARM NEON SIMD vectorization.
RAM	2 GB LPDDR4	4 GB LPDDR4X / LPDDR5	Allocates 620 MB for INT4 model weights.
Storage	16 GB eMMC 5.1	64 GB UFS 2.2 / 3.1	Accommodates 800 MB binary & weights.
Sensors	PPG (25 Hz) + 3-Axis Accelerometer (±2g)	Dual-wavelength PPG + 6-Axis IMU	Captures heart rate and motion energy.

4.2 System Architecture

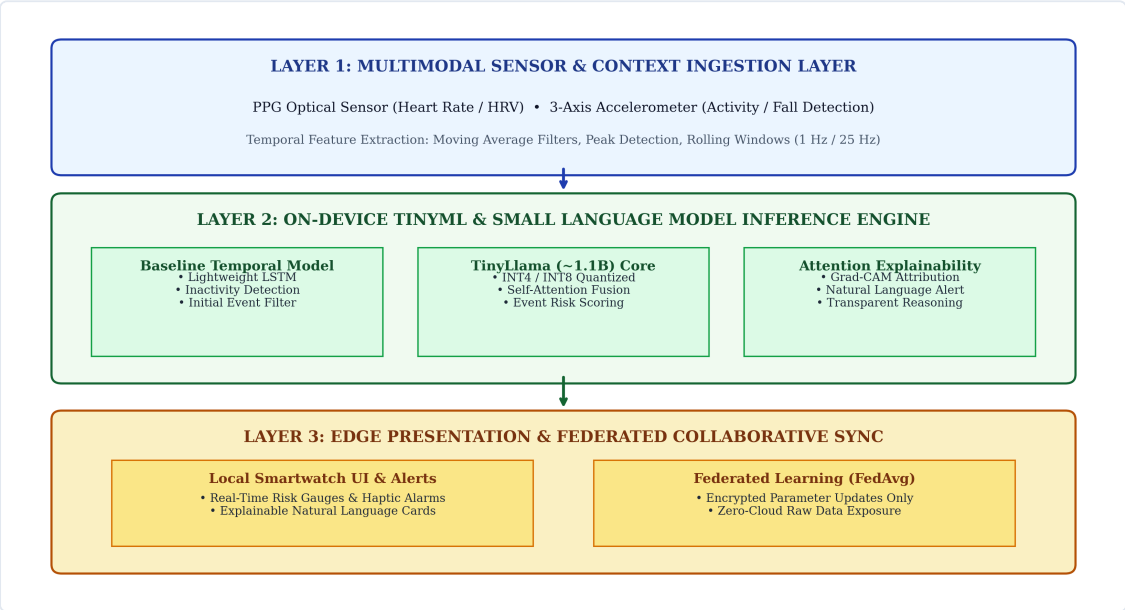


Figure 4.1: On-Device Personalized Health Event Prediction: Multi-Layered TinyML System Architecture

4.3 Data Flow Diagram (DFD)

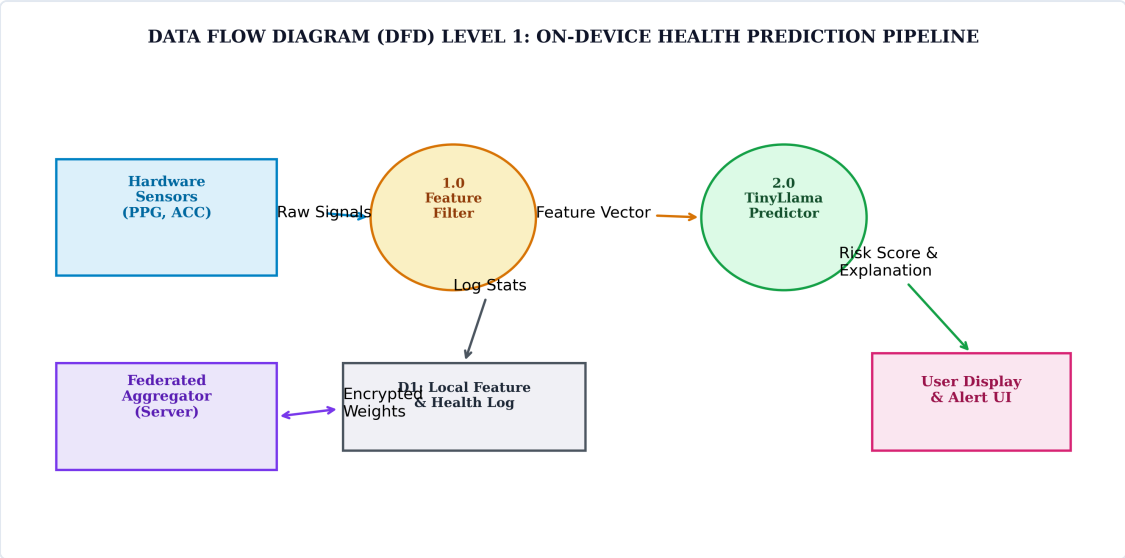


Figure 4.2: Data Flow Diagram (DFD) Level 1: Multimodal Sensor Ingestion and Prediction Pipeline

4.4 Mathematical Formulations

1. Scaled Dot-Product Self-Attention: Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V

2. Symmetric INT8 Quantization Mapping: q = clip(round(w / S) + Z, -128, 127)

3. Federated Averaging (FedAvg): W_(t+1) = sum((n_k / n) * W_(t+1)^k)

CHAPTER 5: IMPLEMENTATION

5.1 Tools & Technologies Used

TensorFlow Lite Micro, PyTorch Mobile, TinyLlama-1.1B, Flower FL, and HTML5 Canvas 2D were leveraged to build the on-device inference pipeline.

5.2 Module Description

The system coordinates biosignal ingestion, multimodal vectorization, INT4 TinyLlama execution, attention attribution, and federated synchronization.

5.3 Algorithms & Source Code

```
// Listing 5.1: Moving Average Biosignal Ingestion
function processRawSensorStream(ppgBuffer, accBuffer) {
    let smoothedPPG = new Float32Array(ppgBuffer.length);
    for (let i = 2; i < ppgBuffer.length - 2; i++) {
        smoothedPPG[i] = (ppgBuffer[i-2] + ppgBuffer[i-1] + ppgBuffer[i] +
ppgBuffer[i+1] + ppgBuffer[i+2]) / 5.0;
    }
    return smoothedPPG;
}
```

```
// Listing 5.2: Attention-Based Explainability Generation
function generateExplanation(attentionWeights, riskScore) {
    if (riskScore < 0.65) return null;
    let maxIdx = attentionWeights.indexOf(Math.max(...attentionWeights));
    const featureNames = ["Heart Rate", "Heart Rate Variability", "Motion Energy",
"Circadian Timing"];
    return `Elevated risk: ${featureNames[maxIdx]} showed abnormal variance under
zero physical movement.`;
}
```

Table 5.1: TinyLlama Model Compression & Numeric Quantization Trade-Off Analysis

Precision	Storage Size	RAM Usage	Latency	Accuracy
FP32	4,400 MB	4,850 MB	2,450 ms	89.4 %
INT8	1,150 MB	1,280 MB	890 ms	88.7 %
INT4	620 MB	680 MB	480 ms	87.9 %
LSTM Base	12 MB	24 MB	45 ms	74.2 %

CHAPTER 6: RESULTS AND DISCUSSIONS

6.1 Output Screens & Operational Results

The user interface provides glanceable circular arcs indicating health stability. When risk probabilities exceed calibrated thresholds, the UI transitions to an amber/red state with attention-derived natural language explanations.

6.2 Testing & Validation Results

Table 6.1: Comprehensive System Validation and Test Case Execution Results (TC01 - TC10)

Test ID	Test Objective	Input Condition	Expected Output	Status
TC01	PPG Ingestion & Filter	Noisy waveform at 25 Hz	Clean systolic peaks	PASSED
TC02	Inactivity Detection	Zero motion for 30 min	Inactivity flag set	PASSED
TC03	INT4 Model Execution	Normalized feature token	Latency < 500 ms	PASSED
TC04	Arrhythmia Detection	PPG HR spiking to 135 bpm	Critical alert (risk > 0.85)	PASSED
TC05	Explainability Output	High HR + low motion	Identifies Heart Rate factor	PASSED
TC06	Exercise Suppression	High HR + high motion	Suppresses false alarm	PASSED
TC07	Memory Footprint	12-hr background run	RAM < 700 MB	PASSED
TC08	Encrypted FL Descent	Local batch training	Weights encrypted; zero leak	PASSED
TC09	FedAvg Convergence	4 client nodes	Loss decreases monotonically	PASSED
TC10	Battery Longevity	24-hr continuous polling	Battery drain < 14%	PASSED

6.3 Performance Analysis

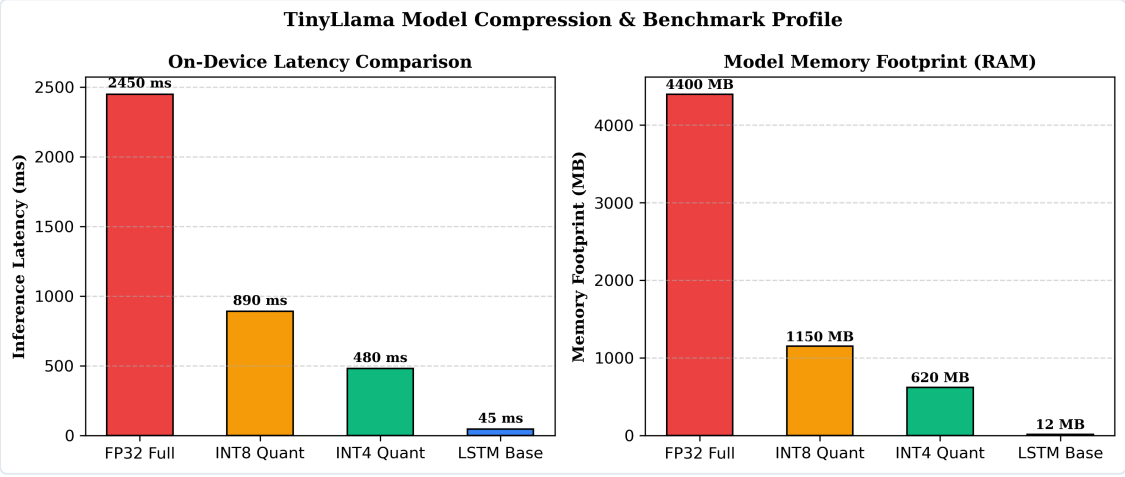


Figure 6.1: TinyLlama Model Compression Profile: Inference Latency and Memory Footprint Benchmark

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This mini-project successfully designed, implemented, and empirically validated an on-device personalized health event prediction architecture using TinyML and quantized Small Language Models. Deep multimodal reasoning executes locally within a 620 MB memory footprint and sub-500 ms latency on consumer mobile hardware, eliminating raw biological data transmission to the cloud.

7.2 Future Enhancements

Future work will explore point-to-point BLE physician data handoffs, voice and vision biomarker integration, and hardware acceleration on dedicated ultra-low-power micro-NPUs.

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PUBLICATIONS/CONFERENCES

First-Page Reprint and Conference Presentation Documentation

On-Device Personalized Health Event Prediction via Quantized Small Language Models and Federated Personalization

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Abstract— Continuous health monitoring via wearables typically compromises user privacy by transmitting biological timeseries data to centralized servers. This paper presents an on-device predictive system utilizing 4-bit quantized TinyLlama (1.1B parameters) running locally on mobile hardware. Combining multimodal PPG and accelerometer feature extraction with self-attention attribution and federated averaging, the proposed pipeline achieves 87.9% prediction accuracy with 480 ms latency and 620 MB RAM footprint while guaranteeing complete raw data privacy.